

Project Testing

Phase 6 of 8 - Project Documentation

Project: OptiCrop: Smart Agricultural Production Optimization Engine

Model Evaluation Results

The trained Logistic Regression model was evaluated on a 20% held-out test split (419 of 2,094 rows remaining after outlier removal):

Metric	Value
Overall Accuracy	94.7%
Training set size	1,675 rows
Test set size	419 rows
Sample prediction: [105, 35, 40, 25, 64, 7, 160]	coffee

Per-Crop Classification Performance (Sample)

Crop	Precision	Recall	F1-Score
rice	1.00	1.00	1.00
maize	0.91	0.91	0.91
chickpea	1.00	1.00	1.00
kidneybeans	0.83	0.82	0.85
mango	0.95	1.00	0.98
coffee	1.00	1.00	1.00

Functional Test Cases

Test Case	Input	Expected Result	Actual Result
Home page loads	GET /	HTTP 200, project intro shown	Pass
About page loads	GET /about	HTTP 200, project description shown	Pass

Test Case	Input	Expected Result	Actual Result
Find Your Crop form loads	GET /findyourcrop	HTTP 200, all 7 input fields present	Pass
Valid prediction	N=105, P=35, K=40, temp=25, 'Best crop for given rainfall is Coffee'	Best crop for given rainfall is Coffee	Pass
Incomplete form submission	Missing required fields	Graceful error, no server crash	Pass

Observations

- The model achieved 94.7% accuracy on the held-out test set, with several crops (rice, chickpea, coffee, apple) achieving perfect or near-perfect precision and recall due to their distinctive environmental signatures.
- Outlier removal on the Phosphorous feature reduced the dataset from 2,200 to 2,094 rows without materially affecting class balance.
- No unhandled exceptions were observed across normal and edge-case input scenarios during test execution.